Welcome to Jupyter Notebook!

In a company's scientific labs, many scientists and engineers keep computational notebooks - 8 1/2" x 13" notebooks with graph paper pages. These notebooks contain an engineer's or scientists handwritten notes containing text, diagrams, and formulas. Jupyter Notebook is a digitized version of this concept. Jupyter Notebook allows you to create and share computational documents containing live code, equations, visualizations/diagrams/images/charts etc., and narrative text.

Each week's lecture material is provided as a Jupyter Notebook. In this tutorial, you'll go through the steps involved in getting started with Jupyter Notebooks. We will look at two platforms:

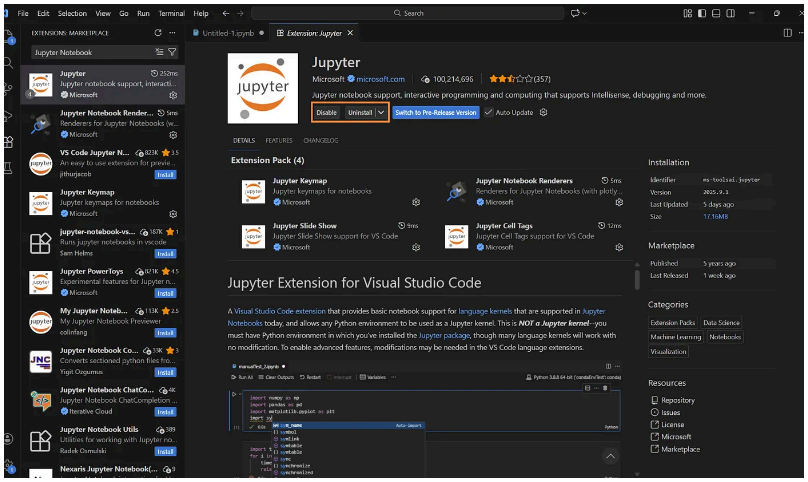
1. Visual Studio Code (VSC)
2. Google Colab

Using Juptyer Notebooks Using VSC

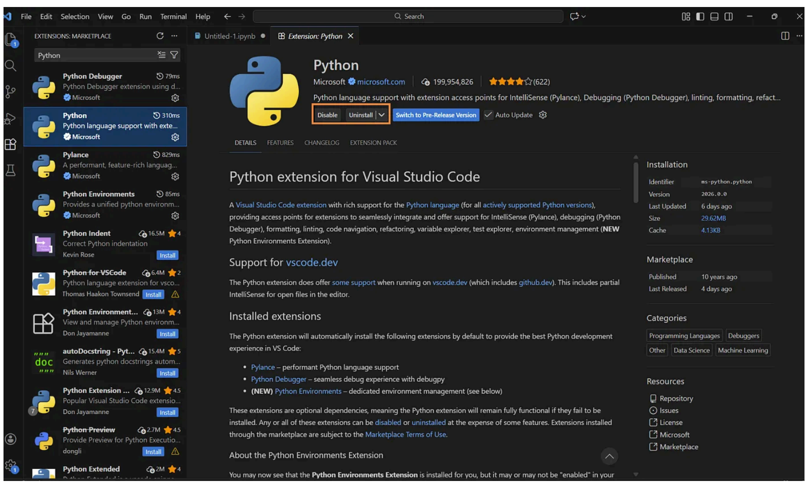
VCS has native support for many file types. Juptyer Notebooks in VSC are basically just another file type. However, most features for file types are provided through extensions.

Installing the Required Extensions

1. Open VSC and and go to the extension view.
2. Search for ‘Jupyter Notebook’ and install the Jupyter Notebook extension.

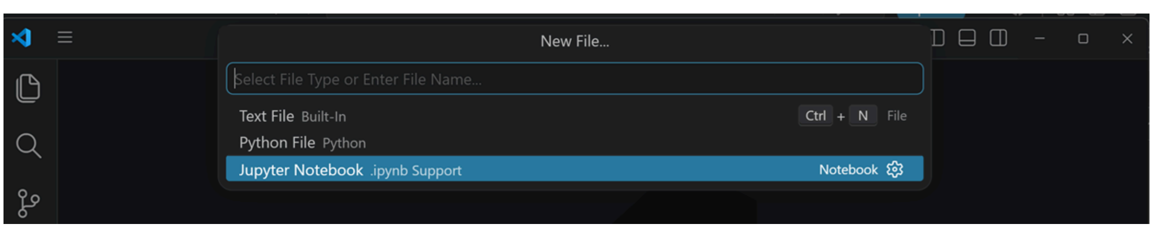


Once done, go back to the extension view and install the Python extension. This is necessary for connecting to the python environment that runs your Notebook code (if you have not already done so).

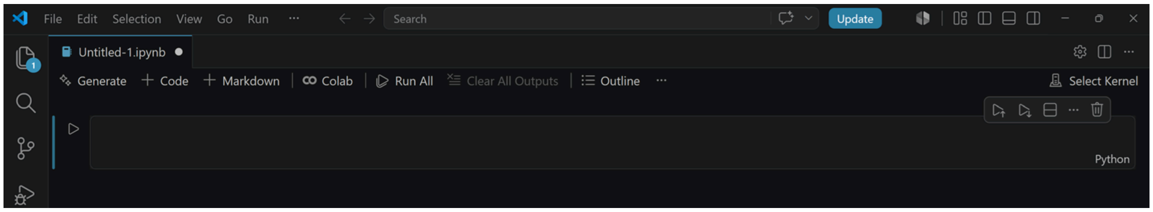


Creating a Notebook

To create a Jupyter Notebook, select File/New File and then select `Jupyter Notebook`:



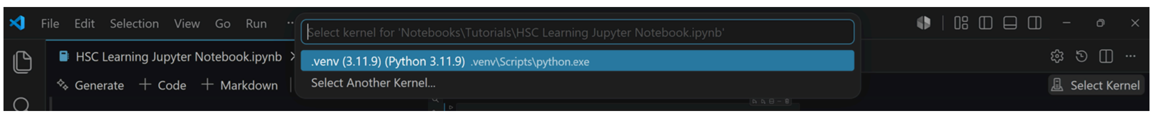
Once selected, a new notebook is created. The first line in the notebook is a `Python` code cell indicated by the word `Python` in the lower right of the cell.



Python Code Cells

In order to write and execute Python code, you have to connect your notebook to a `kernel`. The kernel is basically a “computational engine” that executes the code contained in a Notebook document.

At the top right of your notebook, click on ‘Select Kernel’. You’d see a drop down in the command palette to connect to either Python Environments or Existing Jupyter Server. Select `Python Environments.` Select a Python kernel.



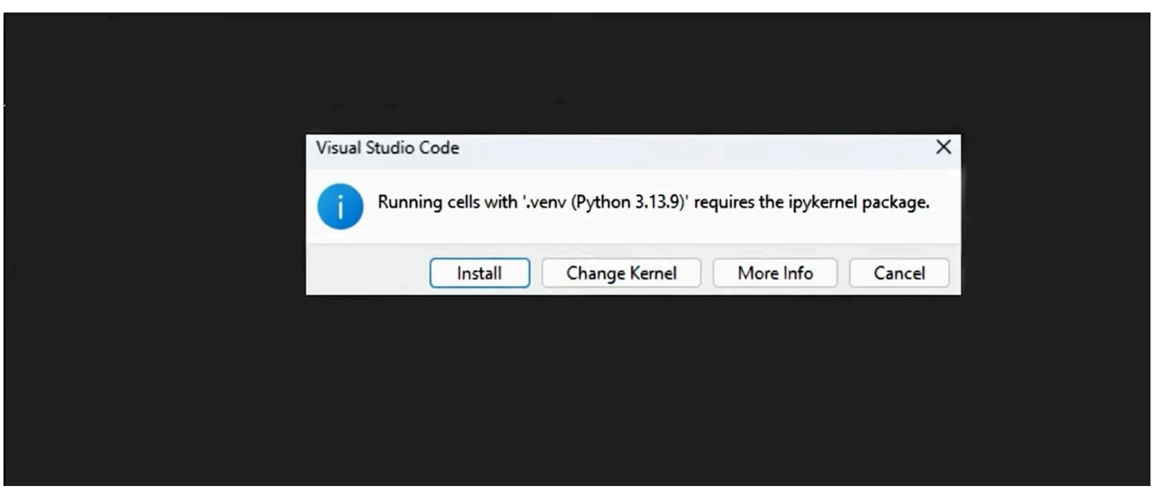
Now let’s run a simple Python code to test this kernel.

In [ ]:
```python
print("Hello, World!")
```

```
Hello, World!
```

Run the code by clicking the play icon on the left side of the cell or by simply entering the Ctrl+Alt+Enter (Windows) or Cmd+Opt+Return (Mac) on your keyboard.

In a situation where the code doesn’t run, you will be prompted to install the `ipykernel` package. The ipykernel package is a lightweight package necessary for running Jupyter Notebook in VS Code.



Once done, the code will run successfully.

Markdown Cells

A `markdown` cell is a cell where you can enter `markdown` text. Markdown is a simple way to format text that looks great on any device. It doesn't do anything fancy like change the font size, color, or type — just the essentials, using keyboard symbols you already know (check out https://commonmark.org/help/ and https://spec.commonmark.org/dingus/ for additional details).

The table below (from the aforementioned websites) is a summary of the symbols required to format text.

Type	Or	... to Get
Italic	_Italic_	*Italic*
Bold	__Bold__	**Bold**
# Heading 1	Heading 1 =========	Heading 1
## Heading 2	Heading 2 ---------	Heading 2
[Link](http://a.com)	[Link][1] ⋮ [1]: http://b.org	Link
`![Image](http://url/a.png)`	`![Image][1]` ⋮ [1]: http://url/b.jpg	
> Blockquote		Blockquote
* List * List * List	- List - List - List	• List • List • List
1. One 2. Two 3. Three	1) One 2) Two 3) Three	1. One 2. Two 3. Three
Horizontal rule: ---	Horizontal rule: ***	Horizontal rule:
`Inline code` with backticks		Inline code with backticks
`` # code block print '3 backticks or' print 'indent 4 spaces' ```	····# code block ····print '3 backticks or' ····print 'indent 4 spaces'	# code block print '3 backticks or' print 'indent 4 spaces'

The alternate to markdown is `plain text` which is supported at least by VSC (when editing a cell, click on the cell type at the lower right to see all the available cell types that VSC supports(ipkernel)). However, using plain text doesn't allow you to *italicize* or to make something **bold**, create headings like this:

Heading 1

or smaller headings like this:

Heading 4

You also cannot make nice block quotes like this:

> Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal...Abraham Lincoln, November 19, 1863

Nor can you create bulleted lists

- First item
- Next Item

or

1. Item 1
2. Item 2

or nice horizontal rules like

---

or inline highlighting like `this`.

None of these formatting types are really complex, but it is much nicer than plain text, and when combined with scientific data and charting and the like, makes for a nice laboratory or scientific report.

Opening a Lecture Notebook

To open a class lecture notebook, simply download it from Caravel (be sure to download the `Jupyter Notebook` file and not the PDF version), and in VSC, select File/Open file or double-click the file to open it in VSC.

Saving a Notebook

To save a notebook, click on File/Save.

Using Jupyter Notebooks Using Google Colab

Colab is a hosted Jupyter Notebook service that requires no setup to use and provides free of charge access to computing resources, including GPUs and TPUs. Colab is especially well suited to machine learning, data science, and education.

To access Colab, go to https://colab.google.com/ If you haven't logged into your Google account, you will be prompted to do so. Once in Colab, click on

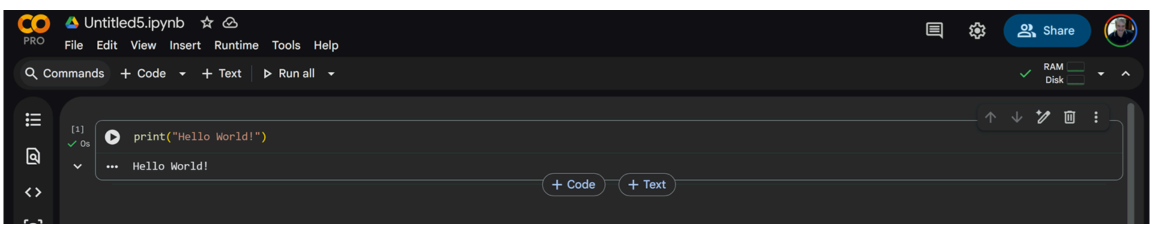
Additional information about Colab is available here: https://research.google.com/colaboratory/faq.html

Creating a Notebook

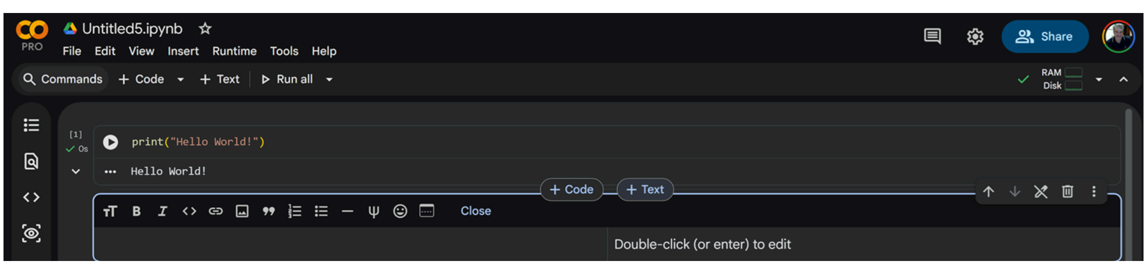
Colab notebooks are stored in Google Drive under your Google account. Once you are in Colab, click on File/New Notebook in Drive.

Markdown Cells

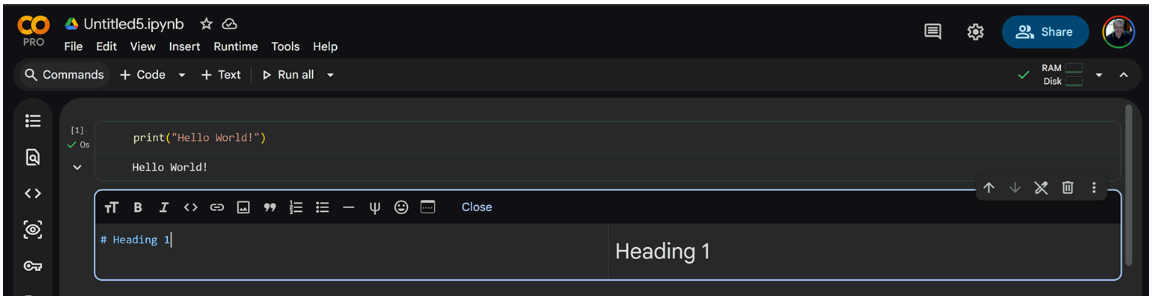
Colab has two cell types: `code` and `text` . To create a markdown cell (known as `text` in Colab), from the menu, select `Insert/Text Cell` . Alternately, you can hover you mouse just above or below an existing cell (in the middle of the cell) to get a pop-up menu and select `text` .



Once you have done that, a text editor will be displayed:

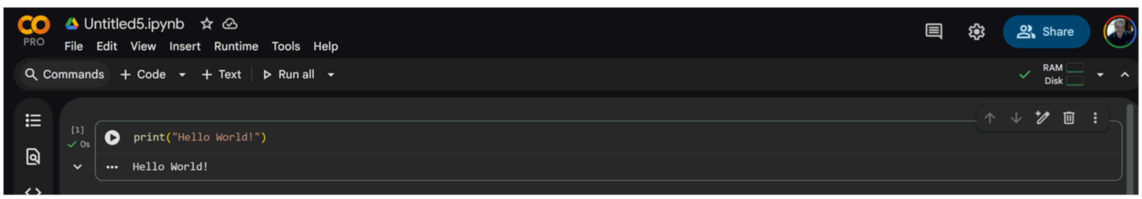


Simply start type using Markdown in the left side of the pane. You will notice that the right-side of the pane will display what the Markdown will look like once you close the cell.



Python Code Cells

To create a Python cell (known as `code` in Colab), from the menu, select `Insert/Code Cell`. Alternately, you can hover you mouse just above or below an existing cell (in the middle of the cell) to get a pop-up menu and select `Code`. Colab can automatically detect the programming language. To execute a coding cell, click on the `play` button like in VSC or Ctrl-Enter.



Opening a Lecture Notebook

To open a lecture or lab notebook, click on the provided link on the course page. This link will automatically open the notebook in Colab in a new browser tab.

Saving a Notebook

There are several ways to save a notebook:

- File/Save (saves a copy to Colab's internal workspace under the /content folder); you will still need to download it to your PC
- File/Save a copy in Drive (if you use Google Drive) (**recommended if a Google Drive user**)
- Download (allows you to save a copy on your PC) (**recommended for all others**)

Lastly, for Google Drive users, Colab saves all notebooks to `My Drive/Colab Notebooks`.

> NOTE: References to "having Google Drive on your PC" is referring to the Google Drive Sync app installed on your PC. Even if you do not have this app installed, as a Google user, you can go to https://drive.google.com to gain access to all your Google Drive files.

Using VSC For Class Materials

If you are an experience VSC user, you can use VCS to clone the course's **Github** public repository.

Before you access any tutorial files, in VSC, navigate to a location on your PC where you store your HSC course files; let's say that is `HSC`. Start VCS and click on "Open Folder." Select this top level folder (or create a different top-level folder to hold this course's files. From the menu, select Terminal/New Terminal. Now type:

`git clone https://github.com/danielgoddu-lang/HSC-CIAI-Course.git`

This will create a subfolder named `HSC-CIAI-Course`. It is your local git repository for the course.

Each week just before class, start VSC, open the `HSC-CIAI-COURSE` subfolder. Next, select Terminal/New Terminal and type:

`git pull`

This will update your local repository with the latest course files.

> WARNING! This Github repository is a public repository. **Do not add or push any files from your local repository to Github!** You will submit your homework using Caravel and not Github.

Submitting Homework Assignments

All assigned labs use Jupyter Notebook. You will be required to download the lab notebook file, modify and execute cells, save (and download) the notebook, and then upload it into the lab assignment for grading and credit.